

Publication List

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1. JH Richter, E Joseph, MC Arcodia, J Berner, JL Demuth, P Falloon, GS Romine, JT Cohen, J Gonzalez-Cruz, M El Gharamti, B Hoppe, S Kumar, A Mariotti, D Mishra, K Pegion, Z Pu, KA Quagraine, KT Quagraine, C Roychoudhury, J Ryan, Z Stone, D Das, B Gaubert, S Kapnick, C Zarzycki. 2026. **Earth System Predictability Across Time Scales for a Resilient Society: A Research Community Perspective.** *Bulletin of the American Meteorological Society*.
2. C Roychoudhury, C He, R Kumar, and AF Arellano. **Revisiting Black Carbon Emission Estimates in the Third Pole: A Hierarchical Bayesian Synthesis Inversion Perspective.** *Environmental Research Communications*. 2026. (Accepted).
3. JM McKinnon, C Roychoudhury, and AF Arellano. **Spatio-temporal pattern analysis of MOPITT total column CO using varimax rotation and singular spectrum analysis.** 2025. *Atmospheric Measurement Techniques*. (Accepted).
4. C Roychoudhury, C He, R Kumar, AF Arellano. 2025. **Diagnosing Aerosol-Meteorological Interactions on Snow within the Earth System: A Proof-of-Concept Study over High Mountain Asia.** *Earth System Dynamics*.
5. C Roychoudhury, C He, R Kumar, JM McKinnon, and AF Arellano. **On the relevance of aerosols to snow cover variability over High Mountain Asia.** 2022. *Geophysical Research Letters*.
6. R Kumar, C He, C Roychoudhury, W Cheng, N Mizukami, and AF Arellano. **High Mountain Asia 12 km Modeled Estimates of Aerosol Transport, Chemistry, and Deposition Reanalysis, 2003–2019. (HMA2_MATCHA, Version 1).** 2024. [Data Set]. *NASA National Snow and Ice Data Center Distributed Active Archive Center*.
7. K Mottungan, C Roychoudhury, V Brocchi, B Gaubert, W Tang, MA Mirrezaei, JM McKinnon, Y Guo, DWT Griffith, DG Feist, I Morino, MK Sha, MK Dubey, M De Mazière, NM Deutscher, PO Wennberg, R Sussmann, R Kivi, TY Goo, VA Velasco, W Wang, and AF Arellano. **Local and regional enhancements of CH₄, CO, and CO₂ inferred from TCCON column measurements.** 2024. *Atmospheric Measurement Techniques*.
8. M Greenslade, Y Guo, G Betito, MA Mirrezaei, C Roychoudhury, AF Arellano, A Sorooshian. **On ozone's weekly cycle for different seasons in Arizona.** 2024. *Atmospheric Environment*.
9. MA Mirrezaei, AF Arellano, Y Guo, C Roychoudhury, A Sorooshian. **Ozone Production over Arid Regions: Insights into Meteorological and Chemical Drivers.** 2024. *Environmental Research Communications*.
10. Y Guo, C Roychoudhury, MA Mirrezaei, R Kumar, A Sorooshian, and AF Arellano. **Investigating Ground-Level Ozone Pollution in Semi-Arid and Arid Regions of Arizona Using WRF-Chem v4.4 Modeling.** 2024. *Geoscientific Model Development*.
11. A Sorooshian, AF Arellano, MP Fraser, P Herckes, G Betito, E Betterton, R Braun, Y Guo, MA Mirrezaei, C Roychoudhury. **Ozone in the Desert Southwest of the United States: A Synthesis of Past Work and Steps Ahead.** 2024. *ACS ES&T Air*.
12. D Das, S Chiao, C Roychoudhury, F Khan, S Chaudhuri, S Mukherjee. **Tropical Cyclone Energy Variability in North Indian Ocean: Insights from ENSO.** 2023. *Climate*.